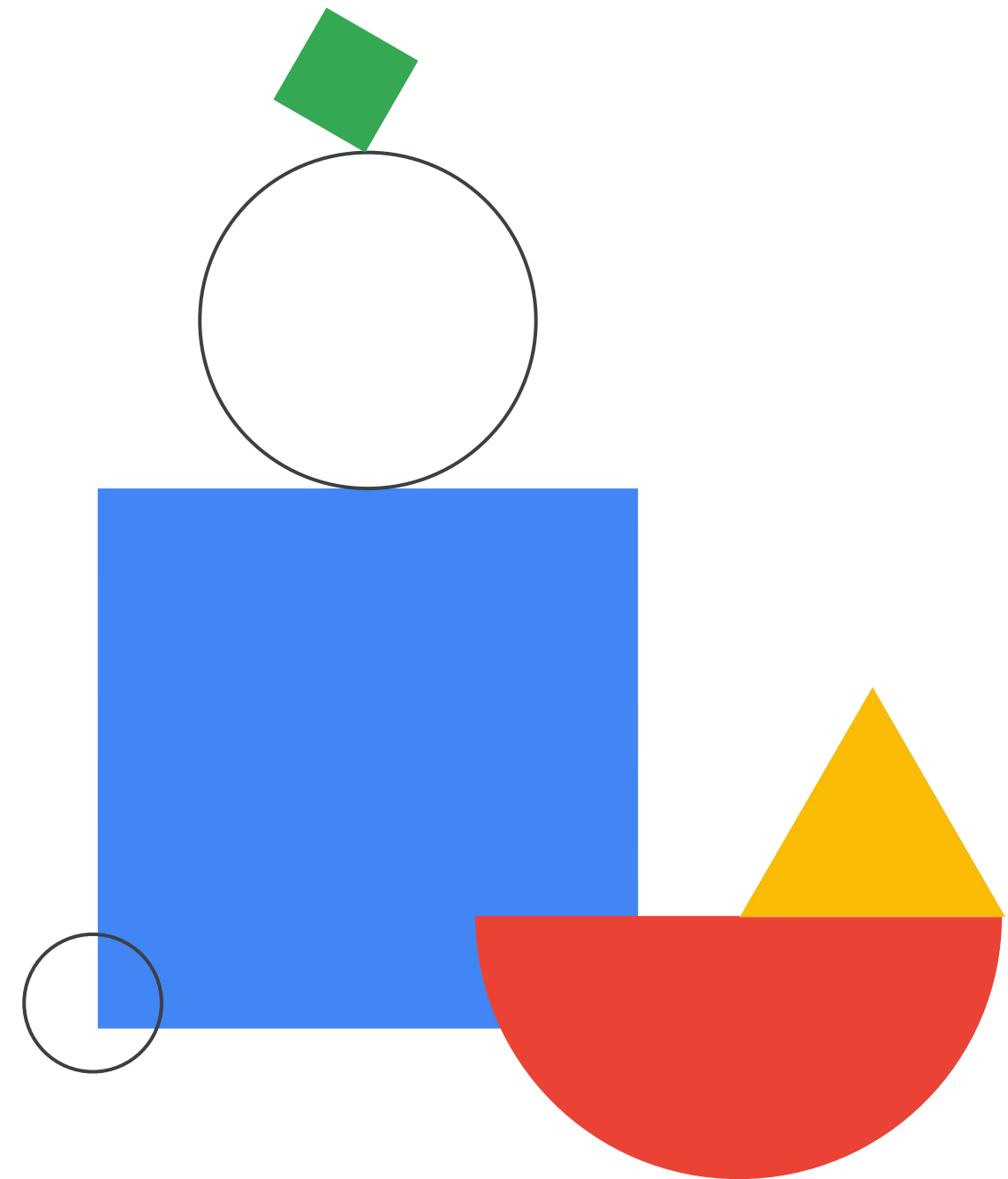
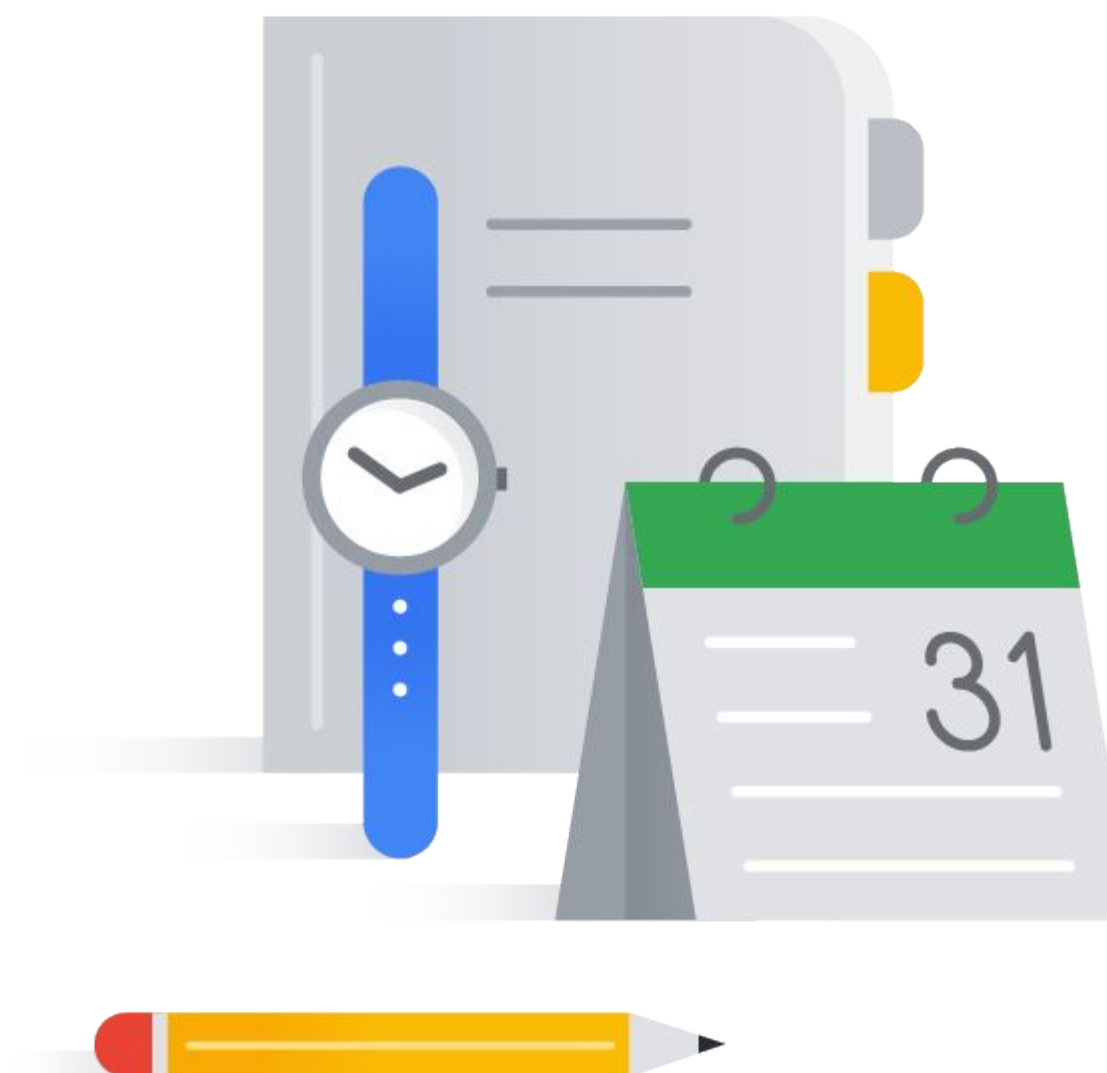


Managed Services



Agenda

- | | |
|----|----------|
| 01 | BigQuery |
| 02 | Dataflow |
| 03 | Dataprep |
| 04 | Dataproc |





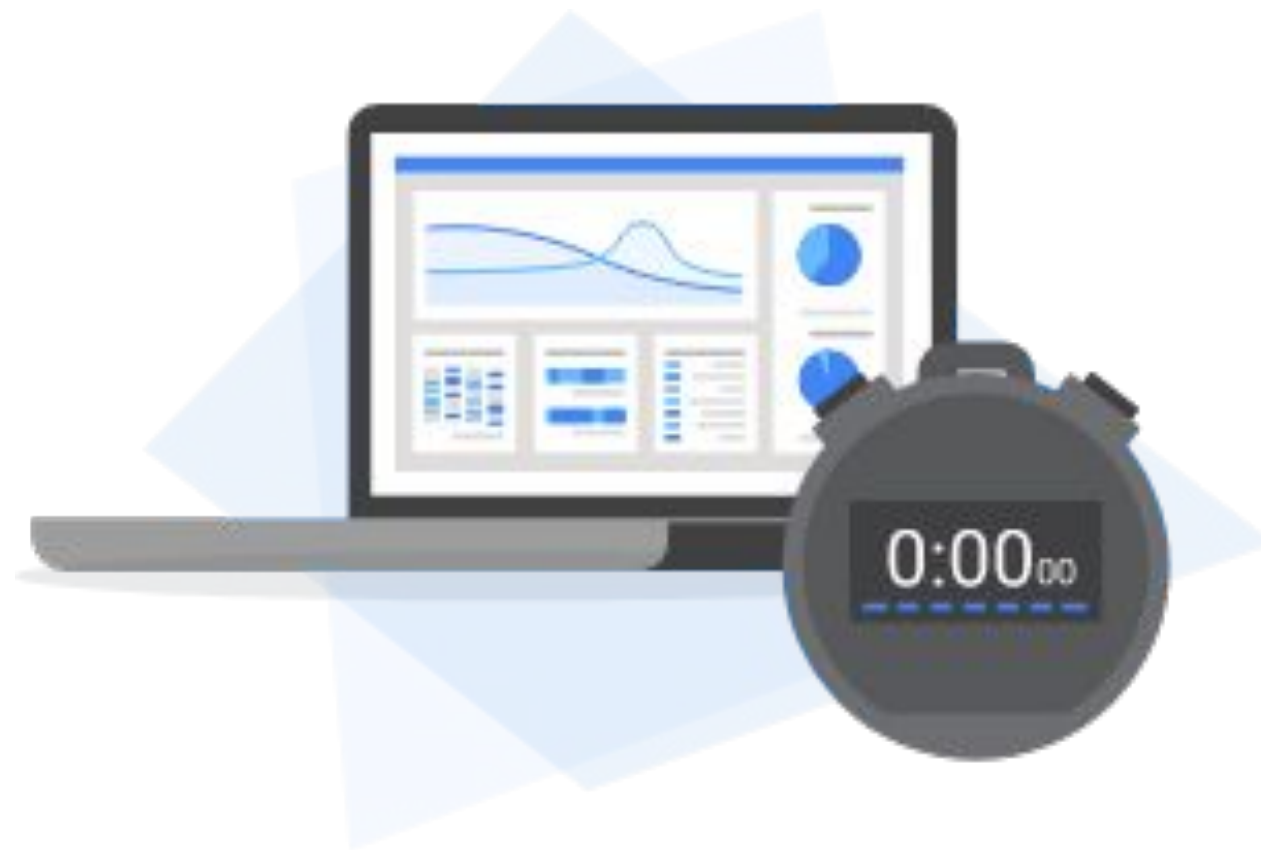
BigQuery

BigQuery is Google Cloud's serverless, highly scalable, and cost-effective cloud data warehouse



BigQuery

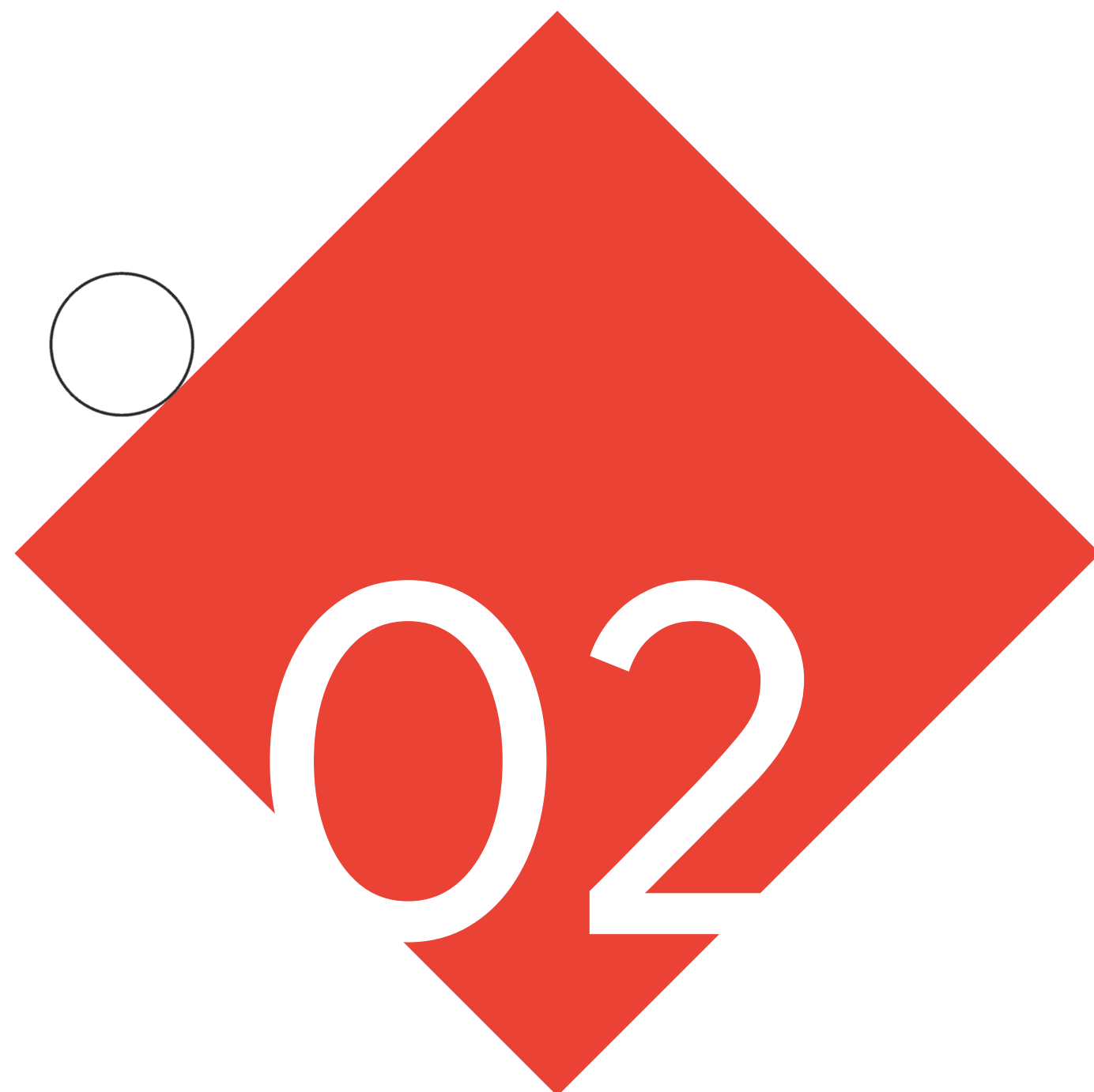
- Fully managed
- Petabyte scale
- SQL interface
- Very fast



Query example

```
WITH groceries AS
  (SELECT "milk" AS dairy,
    "eggs" AS protein,
    "bread" AS grain)
SELECT g.*
FROM groceries AS g;
```

```
+-----+-----+-----+
| dairy | protein | grain |
+-----+-----+-----+
| milk  | eggs    | bread |
+-----+-----+-----+
```



Dataflow

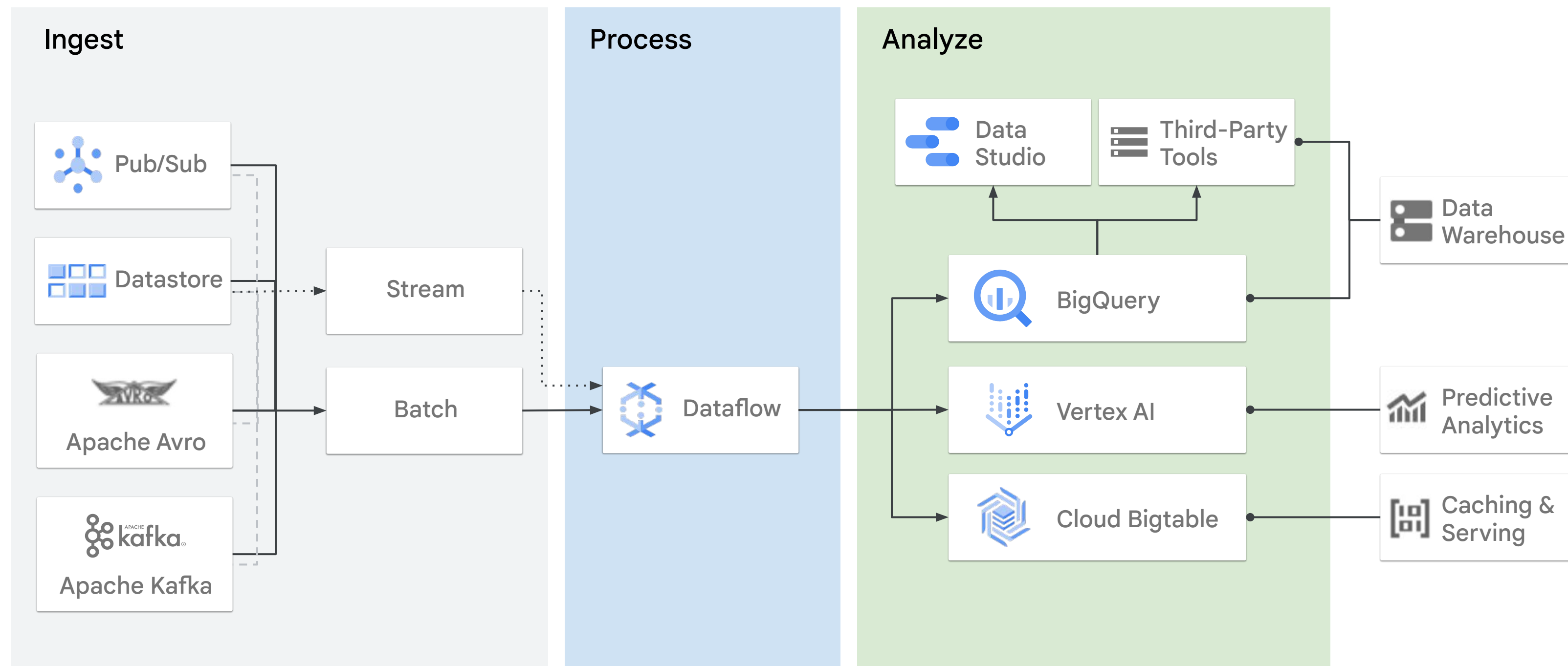
Use Dataflow to execute a wide variety of data processing patterns

- Serverless, fully managed data processing
- Batch and stream processing with autoscale
- Open source programming using  beam
- Intelligently scale to millions of QPS



Dataflow

Data transformation with Dataflow



03



Dataprep

Use Dataprep to visually explore, clean, and prepare data for analysis and machine learning



Dataprep

- Serverless, works at any scale
- Suggests ideal data transformation
- Focus on data analysis
- Integrated partner service operated by Trifacta

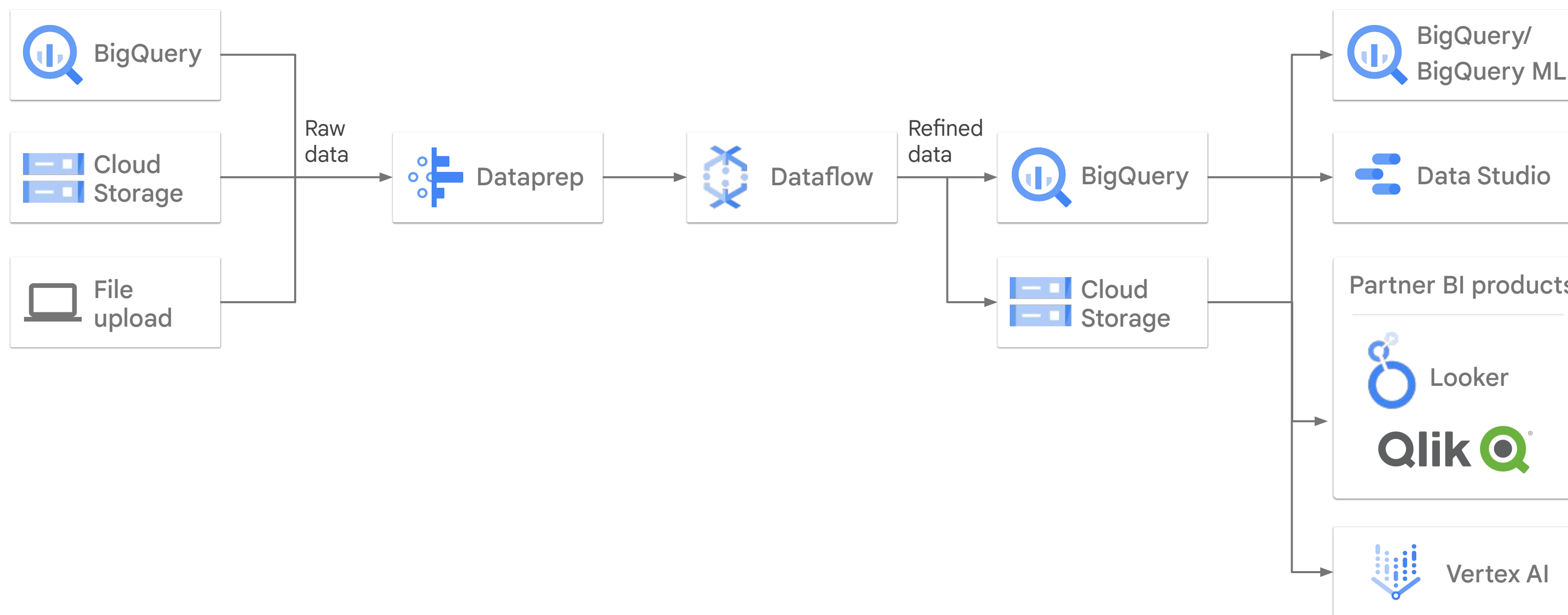


Dataprep architecture

1 Ingestion

2 Preparation & Storage

3 Analysis & ML





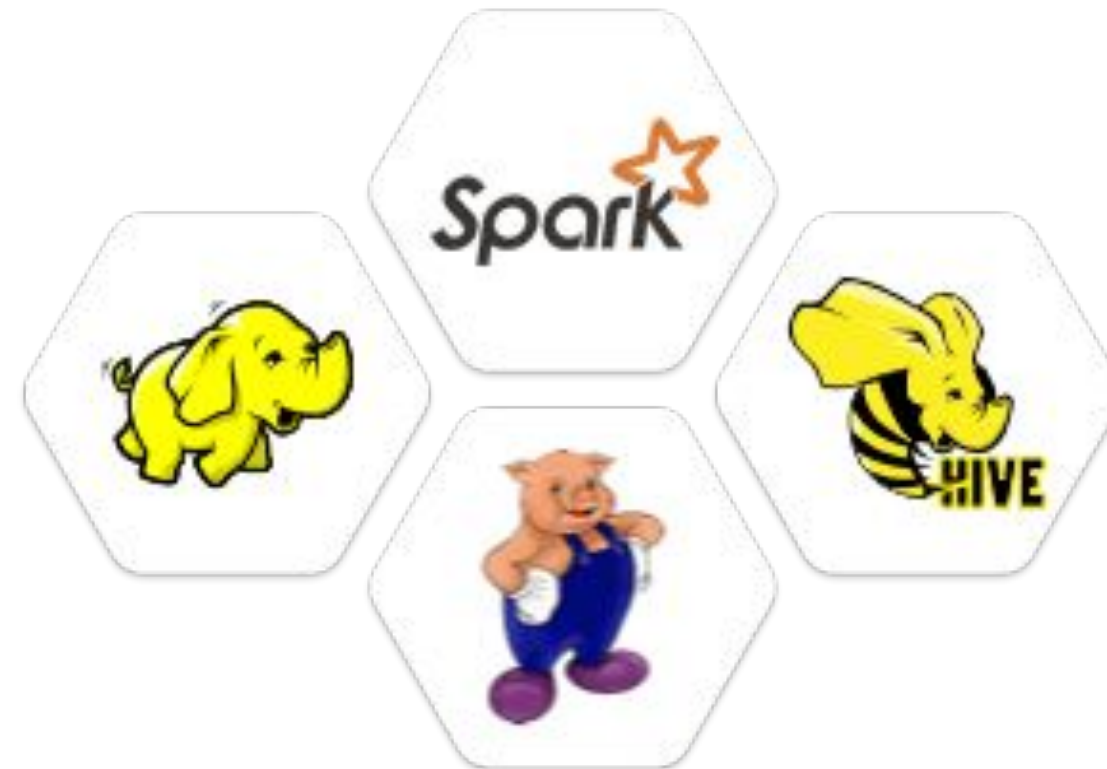
Dataproc

Dataproc is a service for running Apache Spark and Apache Hadoop clusters

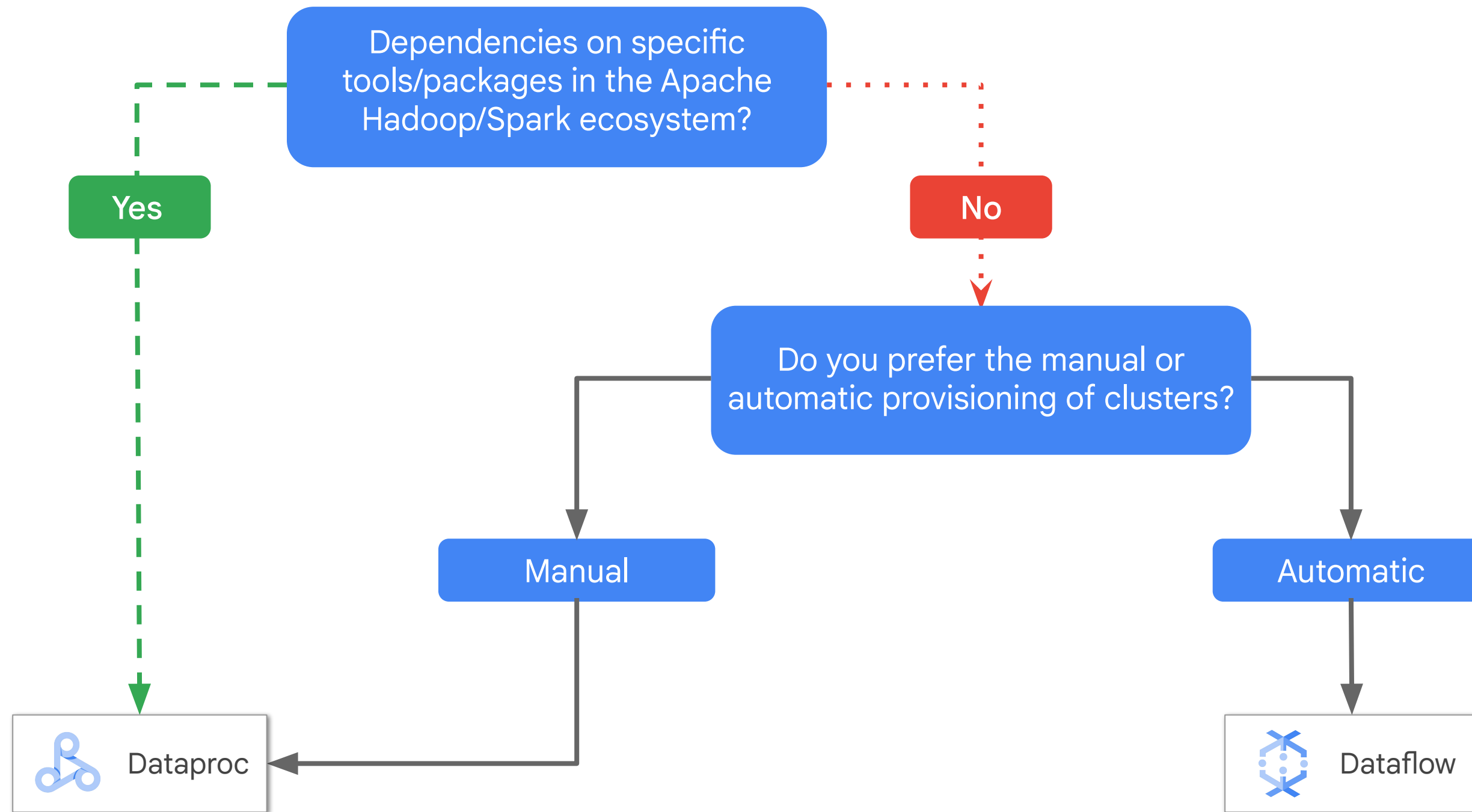
- Low cost (per-second, preemptible)
- Super fast to start, scale, and shut down
- Integrated with Google Cloud
- Managed service
- Simple and familiar



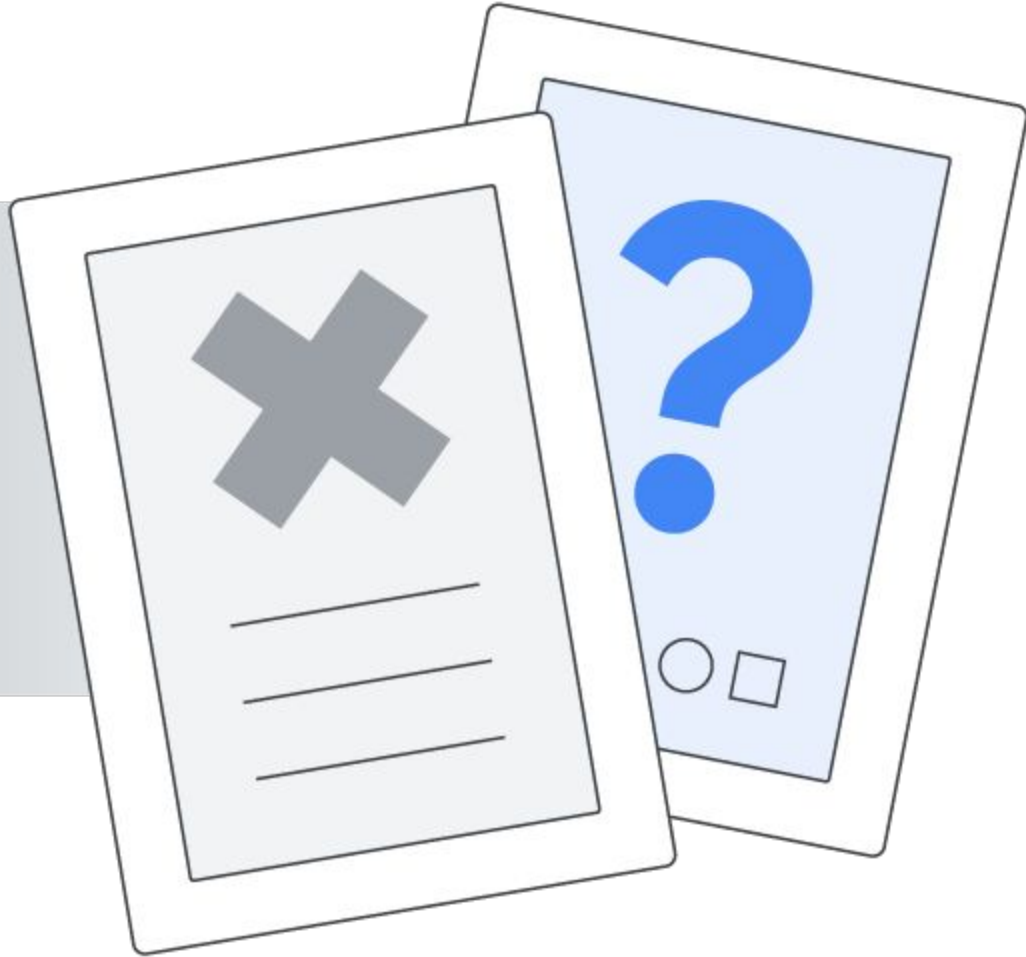
Dataproc



Dataflow versus Dataproc



Quiz



Question #1

Question

How are Managed Services useful?

- A. Managed Services are more customizable than infrastructure solutions
- B. Managed Services may be an alternative to creating and managing infrastructure solutions
- C. If you have an existing infrastructure service, Google will manage it for you if you purchase a Managed Services contract
- D. Managed Services are pay services offered by 3rd party vendors

Question #1

Answer

How are Managed Services useful?

- A. Managed Services are more customizable than infrastructure solutions
- B. Managed Services may be an alternative to creating and managing infrastructure solutions
- C. If you have an existing infrastructure service, Google will manage it for you if you purchase a Managed Services contract
- D. Managed Services are pay services offered by 3rd party vendors



Question #2

Question

Which of the following is a feature of Dataproc?

- A. It typically takes less than 90 seconds to start a cluster
- B. Dataproc allows full control over HDFS advanced settings
- C. Dataproc billing occurs in 10-hour intervals
- D. It doesn't integrate with Cloud Monitoring, but it has its own monitoring system

Question #2

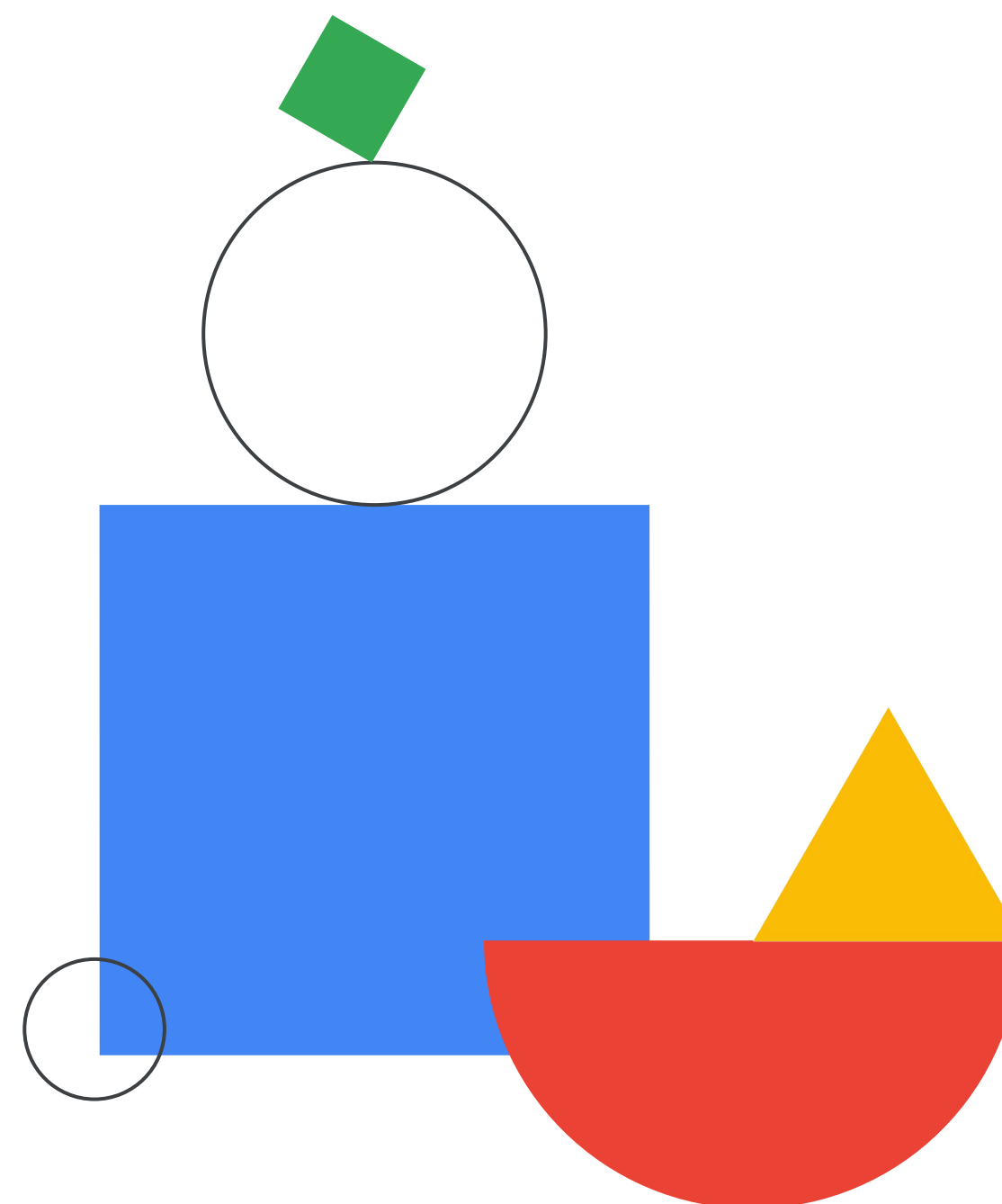
Answer

Which of the following is a feature of Dataproc?

- A. It typically takes less than 90 seconds to start a cluster
- B. Dataproc allows full control over HDFS advanced settings
- C. Dataproc billing occurs in 10-hour intervals
- D. It doesn't integrate with Cloud Monitoring, but it has its own monitoring system

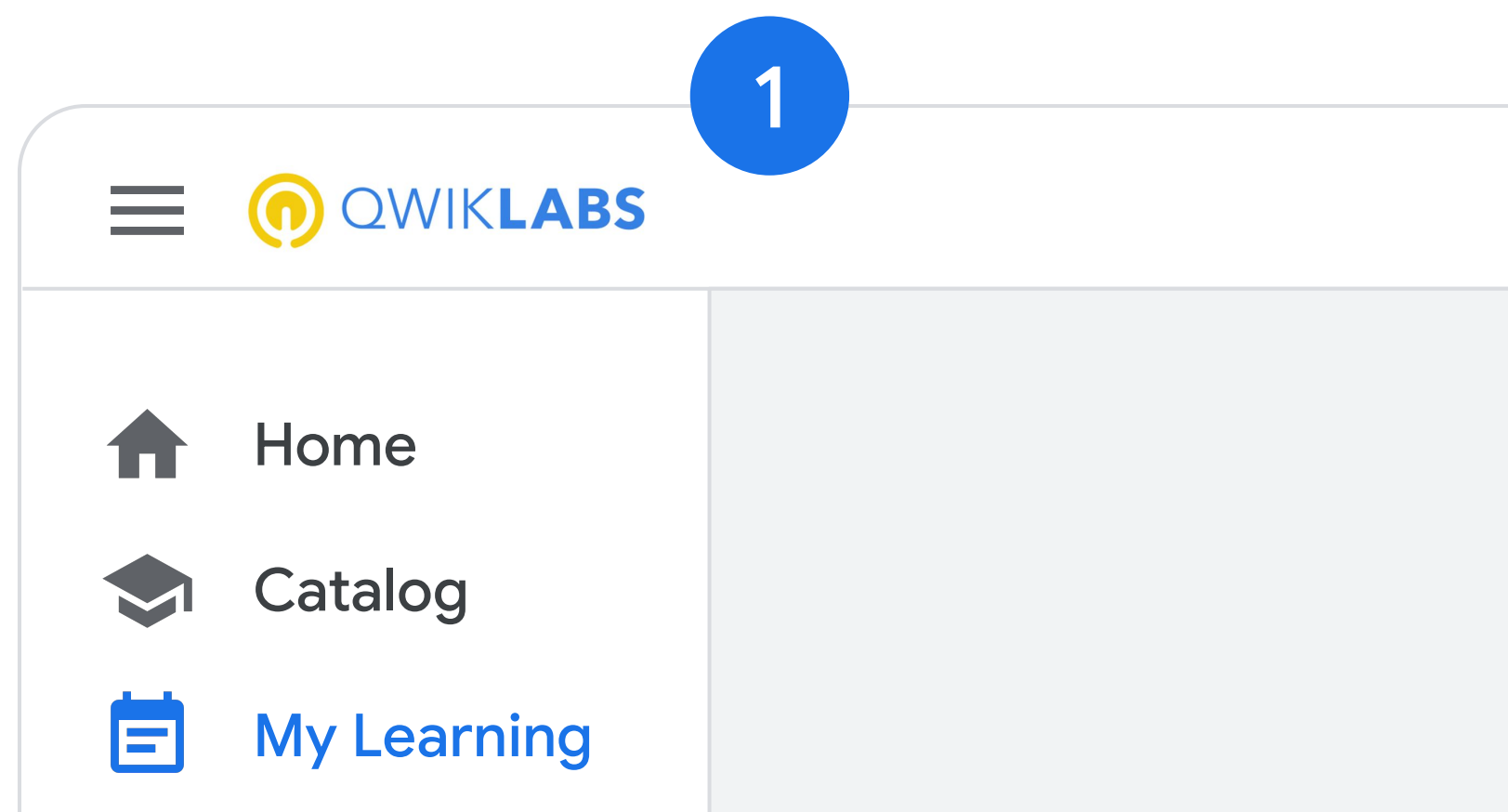


Review: Managed Services

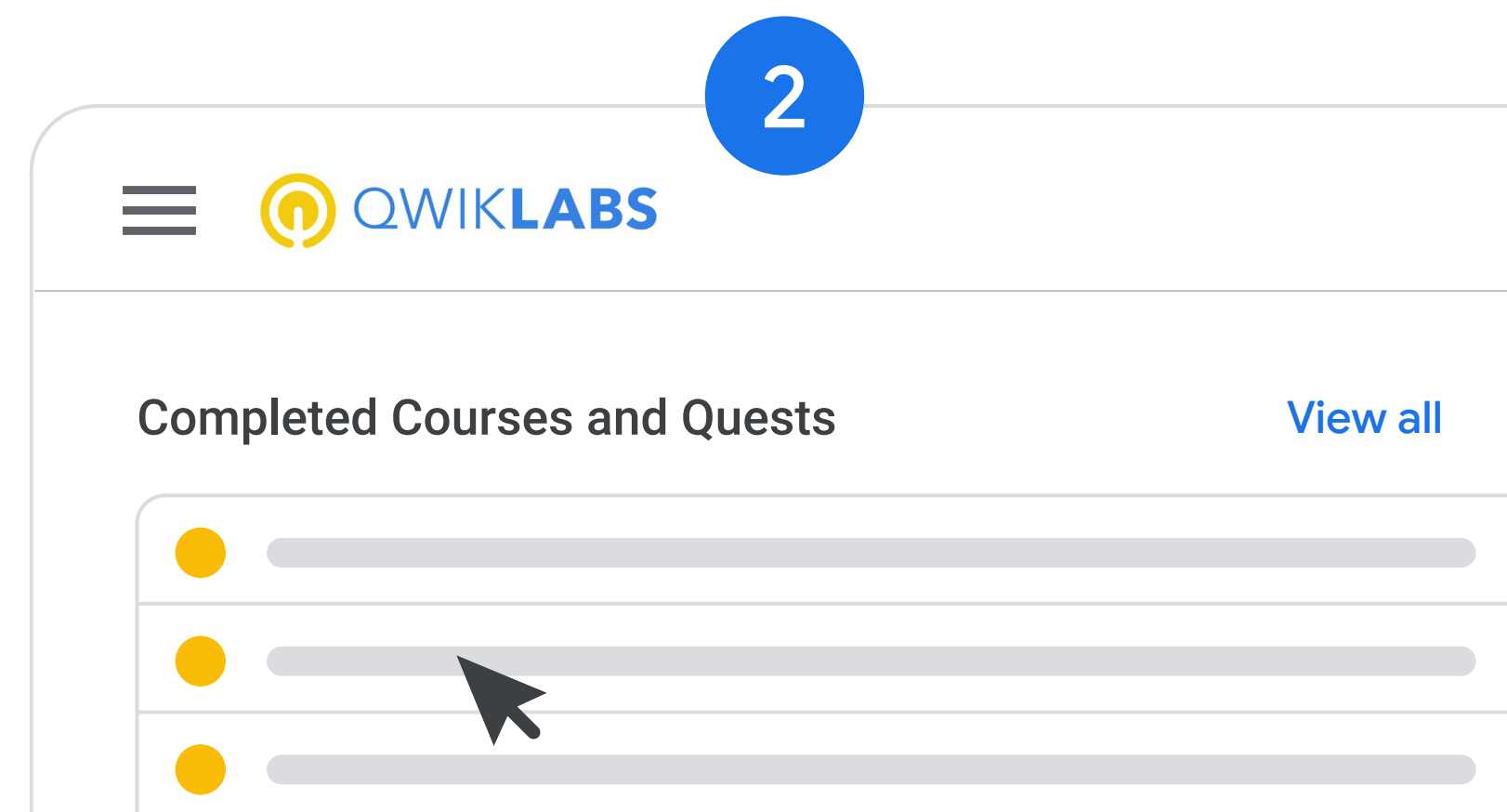


End of class - Materials

Materials are available for 2 years



Click on My Learning in the left-hand navigation bar



Select the class from the **Completed Courses** list

Cloud Infrastructure learning path



Cloud Infrastructure

.....

1

Google Cloud Fundamentals:
Core Infrastructure

2

Architecting with Google
Compute Engine

3

Architecting with Google Cloud:
Design and Process

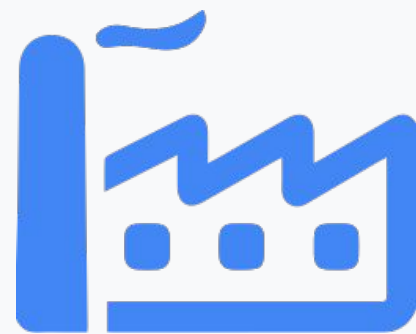
Associate Cloud Engineer Certification



Credibility



Industry
Recognition



Career
Advancement



Personal
Development



Courses in the Cloud Engineer learning path

01 Preparing for your Associate Cloud Engineer Journey ☐

02 Google Cloud Fundamentals: Core Infrastructure ☐

03 Architecting with Google Compute Engine ☒

04 Getting Started with Google Kubernetes Engine ☐

05 Logging, Monitoring and Observability in Google Cloud ☐

06 Getting Started with Terraform for Google Cloud ☐

Skill Badges in the Cloud Engineer learning path

01 Implement Load Balancing on Compute Engine



02 Set Up an App Dev Environment on Google Cloud



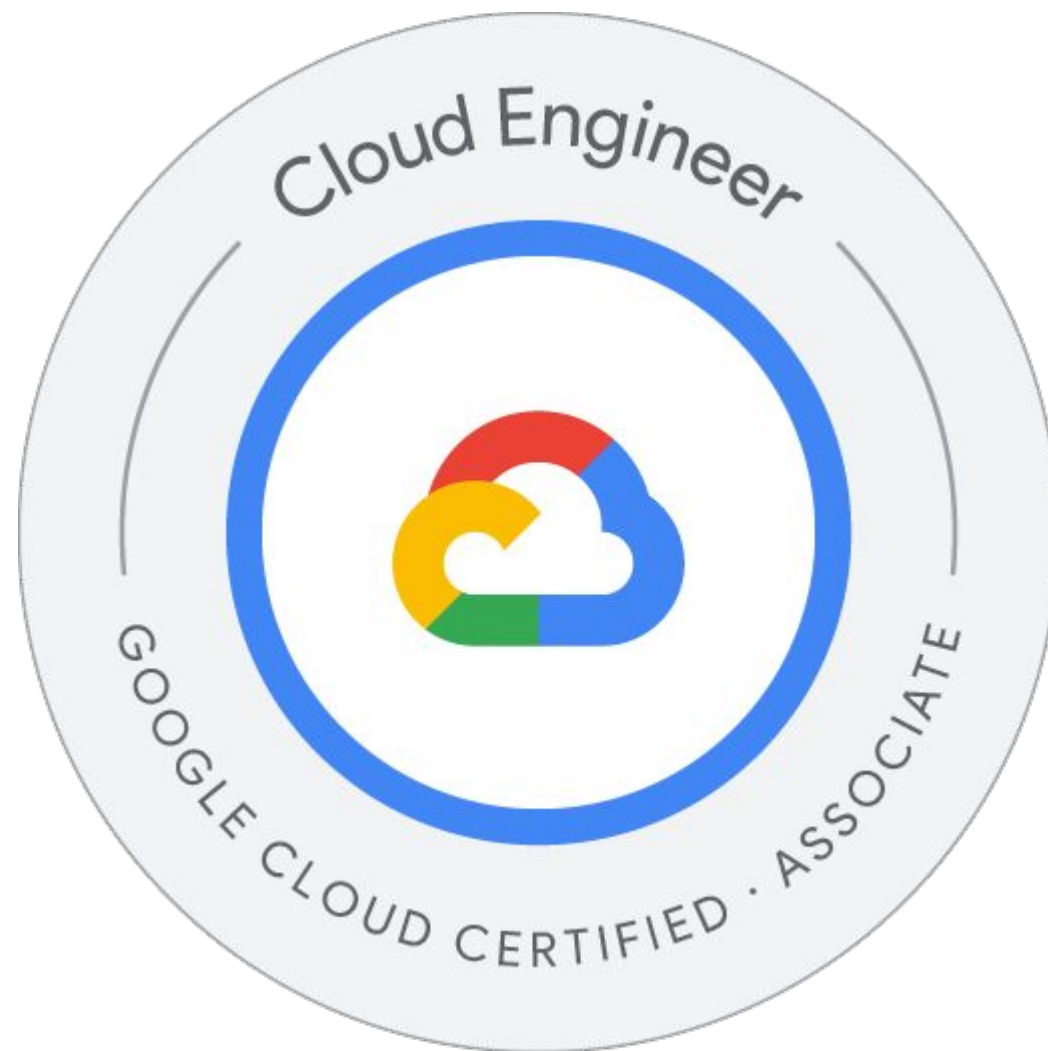
03 Develop your Google Cloud Network



04 Build Infrastructure with Terraform on Google Cloud



Associate Cloud Engineer Certification



Google Cloud

Official Google Cloud Certified

Associate Cloud Engineer Study Guide



Includes interactive online learning environment and study tools:

- 2 custom practice exams
- More than 100 electronic flashcards
- Searchable key term glossary

DAN SULLIVAN

